

OT Infrastructure Monitoring Report

ANALYST
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ORGANIZATION
Green Energy Solutions

ENVIRONMENT
SCADA / ICS Network

OBJECTIVE

The goal of this project was to implement a **passive** network monitoring solution for the SCADA environment at Green Energy Solutions. Given the extreme sensitivity of Industrial Control Systems (ICS), the requirement was to identify device communications without active polling or packet injection.

TECHNICAL IMPLEMENTATION

I developed `ot_final.py` using the **PyShark** library. This tool hooks into the network interface to perform deep packet inspection (DPI) on industrial protocols. Specifically, the script identifies:

- **Modbus:** Monitoring PLC and sensor data exchanges.
- **DNP3:** Identifying utility-grade communications between master stations and RTUs.
- **BACnet:** Detecting environmental and building automation traffic.

CODE LOGIC & PERFORMANCE

```
capture = pyshark.LiveCapture(interface=interface)
for packet in capture.sniff_continuously(packet_count=50):
    filter_ot_traffic(packet)
```

The script utilizes a continuous sniffing loop with integrated exception handling to ensure that malformed packets or network bursts do not cause a system crash, maintaining the reliability required for OT environments.

LOGGING & FORENSIC TRAIL

All identified OT communications are recorded in `ot_scan_log.txt`, providing a clear inventory of active IPs and the industrial protocols they utilize. This data is critical for baseline establishment and unauthorized device detection.